

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

29th May 2014

Subject CT8 – Financial Economics

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)**
- i)** Briefly describe three different types of factor that can be used in a multifactor model of asset returns. (3)
 - ii)** Using mean-variance portfolio theory, prove that the efficient frontier becomes a straight line in the presence of a risk-free asset. (3)
- Consider two independent assets, Gold (G) and Estate (E) with Estate being a more risky asset than Gold. The expected return on Gold is 5% per annum with standard deviation of 6% per annum. The expected return on Estate has a 6% spread over the yield of Gold and its standard is twice that for Gold.
- iii)** If only assets G and E are available, calculate the equation of the efficient frontier in expected return–standard deviation space. (3)
- [9]**
- Q. 2)** Let X and Y be two investment portfolios taking values in $[x,y]$ with cumulative probability distribution functions of returns F_x and F_y respectively. Also let the investor's utility function be Z.
- i)** Write down the equation that the function Z satisfies if the investor prefers more to less. (1)
 - ii)** What does it mean for portfolio X to first order stochastically dominate portfolio Y? Explain. (2)
 - iii)** Prove that if portfolio X first order stochastically dominates portfolio Y, then the investor prefers X to Y. (4)
- [7]**
- Q. 3)**
- i)** Consider an American call option on a stock with 10 months to maturity and risk free rate of interest is 9.5% per annum, exercise price is Rs. 350.
- A dividend of Rs.10 is expected after three months from now and again after six months from now. Show that it can never be optimal to exercise the option on either of the two dividend dates. (3)
- ii)** A stock price follows Geometric Brownian Motion with an expected return of 16% and a volatility of 35%. The current price is Rs. 254.
- What is the probability that a European call option on the stock with an exercise price of Rs. 258 and a maturity date in six months will be exercised?
- What is the probability that a European put option on the stock with the same exercise price and maturity will be exercised? (4)
- [7]**

- Q. 4)** A fund manager has a well diversified portfolio that mirrors the performance of Sensex and is worth Rs. 540 cr. The value of Sensex is 18,000 and the portfolio manager would like to buy an insurance against a reduction of more than 10% of the value of the portfolio over the next 6 months. The risk free rate is 6% per annum. The dividend yield on both the portfolio and the index is 3% and the volatility of the index is 30% per annum.
- i)** If the fund manager buys traded European put options how much would the insurance cost? (4)
 - ii)** If the fund manager decides to provide insurance by keeping part of the portfolio in risk free securities, what should be the initial position? (2)
 - iii)** What will the initial position be if he opts to use a nine – month index future? (2)
 - iv)** Explain carefully alternative strategies available involving traded European call options and show that they lead to the same result (4)
- [12]**
- Q. 5)** Let $Y(t,T)$ be the price at time t of a zero-coupon bond paying Rs.1000 at time T , r_t be the short-rate of interest, P be the real world probability measure and Q the risk neutral probability measure.
- i)** Write down two equations for the price of a zero-coupon bond –
 - a)** Using state-price-deflator approach to pricing.
 - b)** Using risk-neutral approach to pricing (2)
 - ii)** State the Stochastic Differential Equation (SDE) of the short rate r_t under Q for the Vasicek model and the general type of process this SDE represents. (3)
 - iii)** Solve the SDE for the short rate r_t from (b). (5)
 - iv)** Deduce the form of the distribution of the zero-coupon bond price under this model. (2)
- [12]**
- Q. 6)**
- i)** Active fund managers attempt to detect exploitable mispricing between securities and continuously shuffle the portfolio. Is this approach justified by Efficient Market Hypothesis? Why? (2)
 - ii)** Mr. X is an Investment Banker and one of his listed clients ABC Limited has reached out to him for raising debt for a major expansion project. Mr. X believes that this expansion can lead to major operating gains for ABC Limited. Mr. X advises his friend to buy the shares of ABC Limited on this basis.
- Describe how this strategy to buy shares would work in strongly efficient, semi-strongly efficient, weakly efficient and inefficient markets. (2)
- [4]**

- Q. 7)** Consider a corporate bond that will return Re1 per bond to an investor at the end of a year provided the borrower does not default during the year. The constant annual probability of default is 3%. Bonds are purchased at par.

Rahul: Holds 100 such bonds from same borrower.

Sachin: Holds 100 such bonds, each from a different borrower. Each borrower defaults (or not) independently of the other borrowers. Assume annual probability of default across all borrowers is same at 3%.

Calculate variance of Investment and 95% Value at Risk for both Rahul and Sachin

[4]

- Q. 8)** Consider a world with only two risky assets, A and B, and a risk-free asset. Stock A has 200 shares outstanding, a price of Rs.3.00, an expected return of 16% and a volatility of 30%. Stock B has 300 shares outstanding, a price per share of Rs.4.00, an expected return of 10% and a volatility of 15%.

The correlation coefficient is 0.4.

Assume CAPM holds.

- i) What is expected return of the market portfolio? (1)
- ii) What is volatility of the market portfolio? (1)
- iii) What is the beta of each stock? (2)
- iv) What is the risk-free rate? (1)
- v) State the limitations of CAPM (2)

[7]

- Q. 9)**
- i) Define martingales in discrete time (2)
 - ii) Define a Poisson process.
Let $N(t)$ be a Poisson process with parameter λ . Define $M(t) = N(t) - \alpha t$, $\alpha > 0$ is a real number. For what value of α is $M(T)$ a martingale? (3)
 - iii) Suppose $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) with mean μ . Show that $Z_n = \sum_{i=1}^n (X_i - \mu)$ is a martingale. (2)

[7]

- Q. 10)** A stock price is currently Rs. 25. It is known that at the end of every month, the stock price either goes up or goes down by Rs.2. The risk free interest rate is 10% per annum with continuous compounding. Suppose S_T is the stock price at the end of two months.
- What is the value of one month European call option with a strike price of Rs.24? (2)
 - Define $X_t = S_t^2$. What is the value of a derivative that pays off X_T at the end of 2nd month? (3)
 - Suppose we devise an option, that pays $Y = [\{(S_T + S_{T-1})/2\} - K]$ at the end of 2nd month if $[\{(S_T + S_{T-1})/2\} - K] > 0$, and zero otherwise. What will be the price of this option for $K = \text{Rs. } 24$? (4)
- [9]**
- Q. 11)** i) Define Weiner process (2)
- Suppose that the price of a given stock, at time t , X_t follows a geometric Brownian motion with parameters μ and σ , i.e.,
- $$dX_t = \sigma^2 X_t dt + \sigma X_t dW_t, t > 0 \text{ where } W_t \text{ is a Wiener process.}$$
- Let $Z_t = 1/X_t, t > 0$
- State Ito's lemma. (2)
 - Use Ito's lemma to find the stochastic differential equation that Z_t follows (4)
 - Find $E(Z_T | Z_0 = 1)$ (2)
- [10]**
- Q. 12)** Suppose that the spread between the yield on a three year zero coupon corporate bond and the yield on a risk free bond is 50 basis points. The corresponding spread for six year bonds is 80 basis points
- What are the possible credit events and the outcomes of such events? (3)
 - Explain the difference between risk neutral and real world default probability. Which is higher and which should be used for 1) valuation 2) scenario analysis and why? (4)
 - What is the risk neutral probability of default between three and six years in a risk neutral world? Assume zero recovery rate (3)
 - Re-compute the risk neutral probability of default assuming a recovery rate of 25% and 30% for the two corporate bonds. (2)
- [12]**
